

```
html { color: #1a1a1a; background-color: #fdfdfd; } body { margin: 0 auto; max-width: 36em; padding-left: 50px;
padding-right: 50px; padding-top: 50px; padding-bottom: 50px; hyphens: auto; overflow-wrap: break-word; text-rendering:
optimizeLegibility; font-kerning: normal; } @media (max-width: 600px) { body { font-size: 0.9em; padding: 12px; } h1 {
font-size: 1.8em; } } @media print { html { background-color: white; } body { background-color: transparent; color: black;
font-size: 12pt; } p, h2, h3 { orphans: 3; widows: 3; } h2, h3, h4 { page-break-after: avoid; } } p { margin: 1em 0; } a {
color: #1a1a1a; } a:visited { color: #1a1a1a; } img { max-width: 100%; } h1, h2, h3, h4, h5, h6 { margin-top: 1.4em; }
h5, h6 { font-size: 1em; font-style: italic; } h6 { font-weight: normal; } ol, ul { padding-left: 1.7em; margin-top: 1em; } li >
ol, li > ul { margin-top: 0; } blockquote { margin: 1em 0 1em 1.7em; padding-left: 1em; border-left: 2px solid #e6e6e6;
color: #606060; } code { font-family: Menlo, Monaco, Consolas, 'Lucida Console', monospace; font-size: 85%; margin: 0;
hyphens: manual; } pre { margin: 1em 0; overflow: auto; } pre code { padding: 0; overflow: visible; overflow-wrap:
normal; } .sourceCode { background-color: transparent; overflow: visible; } hr { background-color: #1a1a1a; border:
none; height: 1px; margin: 1em 0; } table { margin: 1em 0; border-collapse: collapse; width: 100%; overflow-x: auto;
display: block; font-variant-numeric: lining-nums tabular-nums; } table caption { margin-bottom: 0.75em; } tbody {
margin-top: 0.5em; border-top: 1px solid #1a1a1a; border-bottom: 1px solid #1a1a1a; } th { border-top: 1px solid
#1a1a1a; padding: 0.25em 0.5em 0.25em 0.5em; } td { padding: 0.125em 0.5em 0.25em 0.5em; } header { margin-
bottom: 4em; text-align: center; } #TOC li { list-style: none; } #TOC ul { padding-left: 1.3em; } #TOC > ul { padding-left:
0; } #TOC a:not(:hover) { text-decoration: none; } code{white-space: pre-wrap;} span.smallcaps{font-variant: small-
caps;} div.columns{display: flex; gap: min(4vw, 1.5em);} div.column{flex: auto; overflow-x: auto;} div.hanging-
indent{margin-left: 1.5em; text-indent: -1.5em;} /* The extra [class] is a hack that increases specificity enough to override
a similar rule in reveal.js */ ul.task-list[class]{list-style: none;} ul.task-list li input[type="checkbox"] { font-size: inherit;
width: 0.8em; margin: 0 0.8em 0.2em -1.6em; vertical-align: middle; } .display.math{display: block; text-align: center;
margin: 0.5rem auto;}
```

CareConnect Clinic Appointment System

Complete System Specification Document — Practical Lab Solution

23 July 2026

[Step 1 — Project Charter & Feasibility Study](#)

- [1.1 Project Title](#)
- [1.2 Problem Statement](#)
- [1.3 Project Goals](#)
- [1.4 Scope](#)
- [1.5 Key Stakeholders](#)
- [1.6 Feasibility Analysis](#)

[Step 2 — Requirements Analysis & Data Modeling \(Part 1\)](#)

- [2.1 User Stories](#)
- [2.2 Functional Requirements](#)
- [2.3 Non-Functional Requirements](#)
- [2.4 Data Flow Diagrams](#)

[Step 3 — Requirements Analysis & Data Modeling \(Part 2\)](#)

- [3.1 Entity-Relationship Diagram \(ERD\)](#)
- [3.2 Database Normalization](#)

[Step 4 — System Design & Interface Prototyping](#)

- [4.1 User Interface \(UI\) Design](#)
- [4.2 Database Schema](#)
- [4.3 System Architecture](#)

[Step 5 — System Implementation & Review](#)

- [5.1 System Specification Document \(Consolidated\)](#)
- [5.2 Final Presentation Outline \(10–15 minutes\)](#)

Step 1 — Project Charter & Feasibility Study

1.1 Project Title

CareConnect Clinic Appointment System

1.2 Problem Statement

CareConnect currently relies on paper folders and a physical appointment book to manage patients and scheduling. This manual process creates several recurring problems:

- **Scheduling conflicts** — double-booked time slots because the paper diary cannot be checked in real time by more than one staff member at once.
- **Difficulty retrieving patient records** — paper folders are frequently misplaced, misfiled, or unavailable when a patient is being seen, delaying diagnosis and treatment.
- **Administrative burden** — front-desk staff spend a large share of their day on phone calls, manual filing, and reconciling the appointment book, leaving less time for patient care.
- **Lost patient history** — when folders go missing, doctors lose access to prior visit notes, allergies, and prescriptions.
- **Long patient wait times** — walk-ins, phone bookings, and paper records combine to create queues and unpredictable delays.

1.3 Project Goals

The new system will:

1. Allow patients to **book, reschedule, and cancel appointments online**, 24/7, without calling during business hours.
2. Give doctors and staff **instant, reliable access to patient medical history** at the point of care.
3. **Reduce no-show rates** through automated SMS/email reminders.
4. **Eliminate double-booking** through a single, shared, real-time schedule.
5. **Reduce administrative workload** by automating registration, reminders, and reporting.
6. Provide clinic management with **attendance and doctor-utilization reporting** to support staffing decisions.

1.4 Scope

In scope — the system will manage: - Patient profiles and contact information - Doctor availability and schedules - Appointment booking, rescheduling, and cancellation - Basic visit notes and medical history linked to appointments - Automated reminders and notifications - Attendance and utilization reporting

Out of scope — the system will **not** handle: - Medical billing or insurance claims processing - Pharmacy inventory or e-prescription dispensing to external pharmacies - Telemedicine/video consultation delivery (may be considered in a future phase) - Integration with external hospital records systems (future phase)

1.5 Key Stakeholders

Stakeholder	Interest / Role
Clinic Manager	Owns budget and operational outcomes; needs reporting and oversight
Doctors	Primary clinical users; need fast access to schedules and patient history
Nurses	Support clinical workflow; update vitals/notes ahead of doctor visits
Administrative / Front-desk Staff	Registers patients, manages walk-ins, handles phone bookings
Patients	End users of the online booking portal; primary beneficiaries of reduced wait times
IT/System Administrator	Maintains servers, security, and backups

1.6 Feasibility Analysis

1.6.1 Technical Feasibility

The required technology is mature, widely available, and low-risk:

- **Hardware:** a secure on-site or cloud-hosted server (or a managed cloud instance, e.g. AWS/Azure), front-desk workstations, and tablets for doctors/nurses to review schedules at the bedside.
- **Software:** a relational database management system (e.g. PostgreSQL/MySQL), a web application framework for the booking portal, and an appointment/notification engine (SMS/email gateway, e.g. Twilio or a local SMS provider).
- **Network/Security:** HTTPS/TLS for all traffic, encrypted database storage, and role-based access control.

All of these components are standard, well-documented, and already used successfully by comparable small clinics, so the project is judged **technically feasible** with low risk.

1.6.2 Economic Feasibility — Simple Cost-Benefit Analysis

(Figures in Nigerian Naira, ₦; approximate reference rate used: US\$1 ≈ ₦1,500)

Estimated one-time and recurring costs

Cost Item	Estimate
Development (analysis, design, build, testing — ~12 weeks)	₦27,000,000
Software licenses / cloud hosting (annual)	₦3,600,000/yr
SMS/email notification gateway (annual)	₦900,000/yr
Staff training (one-time)	₦1,500,000
Hardware (2 tablets + minor upgrades)	₦2,250,000
Total Year 1 cost	₦35,250,000
Recurring annual cost (Year 2+)	₦4,500,000

Estimated annual benefits

Benefit Item	Estimate
Reduced administrative staff time (≈10 hrs/week saved × ₦22,500/hr × 52 weeks)	₦11,700,000/yr
Reduced no-shows (fewer wasted appointment slots, estimated 15% reduction)	₦9,000,000/yr
Increased patient throughput (better scheduling utilization)	₦13,500,000/yr
Reduced errors/rework from lost records	₦3,750,000/yr
Total estimated annual benefit	₦37,950,000/yr

Conclusion: Year 1 net benefit ≈ ₦37,950,000 – ₦35,250,000 = **+₦2,700,000**, and every year thereafter the system saves approximately **₦37,950,000 – ₦4,500,000 = ₦33,450,000**. The payback period is under 12 months, so the project is judged **economically feasible**.

1.6.3 Operational Feasibility

Clinic staff already use basic computer applications, so the learning curve for a well-designed booking and records interface is expected to be shallow. Key operational requirements:

- A half-day hands-on training session for administrative staff and a shorter orientation for doctors/nurses, focused on daily workflows (check-in, schedule review, note entry).
- A short **data-privacy and HIPAA/PDPA-style-compliance briefing** covering password hygiene, access-logging, and proper patient-record handling, since the system will centralize sensitive health data.
- A phased rollout (pilot with one doctor's schedule before clinic-wide go-live) to build staff confidence and surface issues early.

Given management support and a structured training plan, the project is judged **operationally feasible**.

Deliverable produced above: *CareConnect Project Charter & Feasibility Study* (Sections 1.1–1.6).

Step 2 — Requirements Analysis & Data Modeling (Part 1)

2.1 User Stories

1. **As a Patient**, I want to book an appointment online so that I don't have to call the clinic during busy hours.
2. **As a Patient**, I want to receive an SMS/email reminder 24 hours before my appointment so that I don't forget and miss my slot.
3. **As a Patient**, I want to view and cancel/reschedule my upcoming appointments so that I can manage changes to my plans without calling in.
4. **As a Patient**, I want to see only the doctor's actually-available time slots so that I don't waste time requesting a slot that is already taken.
5. **As a Doctor**, I want to view my daily schedule and patient history notes before the appointment starts so that I can prepare for each consultation.
6. **As a Doctor**, I want to add visit notes and a diagnosis after a consultation so that the patient's medical history stays current for future visits.
7. **As a Nurse**, I want to record a patient's vitals before the doctor sees them so that the doctor has up-to-date information at the point of care.
8. **As an Administrative Staff member**, I want to register a new patient quickly so that walk-ins can be seen without long delays.
9. **As an Administrative Staff member**, I want to be alerted when a doctor is unavailable (e.g. sick leave) so that I can rebook affected patients before they arrive.
10. **As a Clinic Manager**, I want to view attendance and doctor-utilization reports so that I can make informed staffing and scheduling decisions.

2.2 Functional Requirements

ID	Functional Requirement
FR-1	The system shall allow patients to create an online account and log in securely.
FR-2	The system shall allow patients to search doctor availability and book an appointment.
FR-3	The system shall allow patients and staff to reschedule or cancel an appointment.
FR-4	The system shall prevent two appointments from being booked in the same doctor time slot (double-booking prevention).
FR-5	The system shall allow staff to register new patients and edit patient profiles.
FR-6	The system shall allow doctors/nurses to view daily schedules and patient medical history.
FR-7	The system shall allow doctors to record visit notes, diagnoses, and prescriptions after a consultation.
FR-8	The system shall send automated appointment confirmations and reminders via SMS/email.
FR-9	The system shall allow the clinic manager to generate attendance and doctor-utilization reports.
FR-10	The system shall log all access to patient medical records for audit purposes.

2.3 Non-Functional Requirements

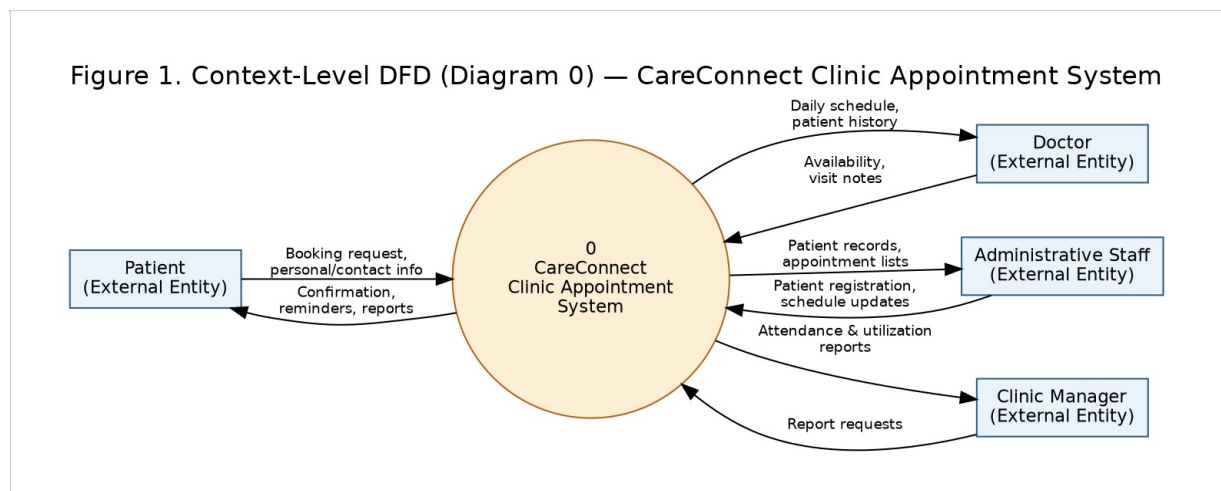
ID	Non-Functional Requirement
NFR-1	Data privacy/compliance: all patient health data must be stored encrypted at rest and in transit, and access must be restricted by role (HIPAA/PDPA-aligned).
NFR-2	Availability: the booking portal must be available 99.5% of the time (excluding scheduled maintenance windows).
NFR-3	Performance: schedule and record lookups must return results

	within 2 seconds under normal load.
NFR-4	Usability: front-desk staff should be able to complete a new patient registration in under 3 minutes after training.
NFR-5	Auditability: every create/update/delete on a medical record must be timestamped and attributed to a user account.
NFR-6	Scalability: the system must support growth to at least 10,000 patient records without redesign.
NFR-7	Backup/Recovery: the database must be backed up daily with a recovery point objective (RPO) of 24 hours.

2.4 Data Flow Diagrams

2.4.1 Context-Level DFD (Diagram 0)

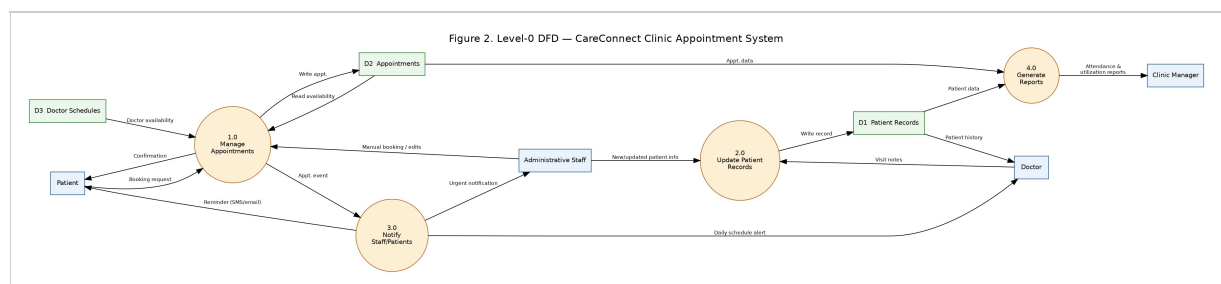
The context diagram treats the entire CareConnect system as a single process and shows its data exchanges with the four external entities: Patient, Doctor, Administrative Staff, and Clinic Manager.



Context-Level DFD

2.4.2 Level-0 DFD

The Level-0 diagram decomposes the single system process into four core subprocesses — **1.0 Manage Appointments**, **2.0 Update Patient Records**, **3.0 Notify Staff/Patients**, and **4.0 Generate Reports** — and shows how they read from and write to three data stores: Patient Records (D1), Appointments (D2), and Doctor Schedules (D3).



Level-0 DFD

Deliverable produced above: user stories, requirements list, and DFDs (Sections 2.1–2.4).

Step 3 — Requirements Analysis & Data Modeling (Part 2)

3.1 Entity-Relationship Diagram (ERD)

3.1.1 Core Entities and Attributes

Entity	Attributes
Patient	PatientID (PK), FirstName, LastName, DOB, Gender, Phone, Email, Address
Doctor	DoctorID (PK), FirstName, LastName, Specialization, Phone, Email
Appointment	AppointmentID (PK), PatientID (FK), DoctorID (FK), AppointmentDate, AppointmentTime, Status, Reason
MedicalRecord	RecordID (PK), PatientID (FK), AppointmentID (FK), DoctorID (FK), VisitDate, Diagnosis, Prescription, Notes

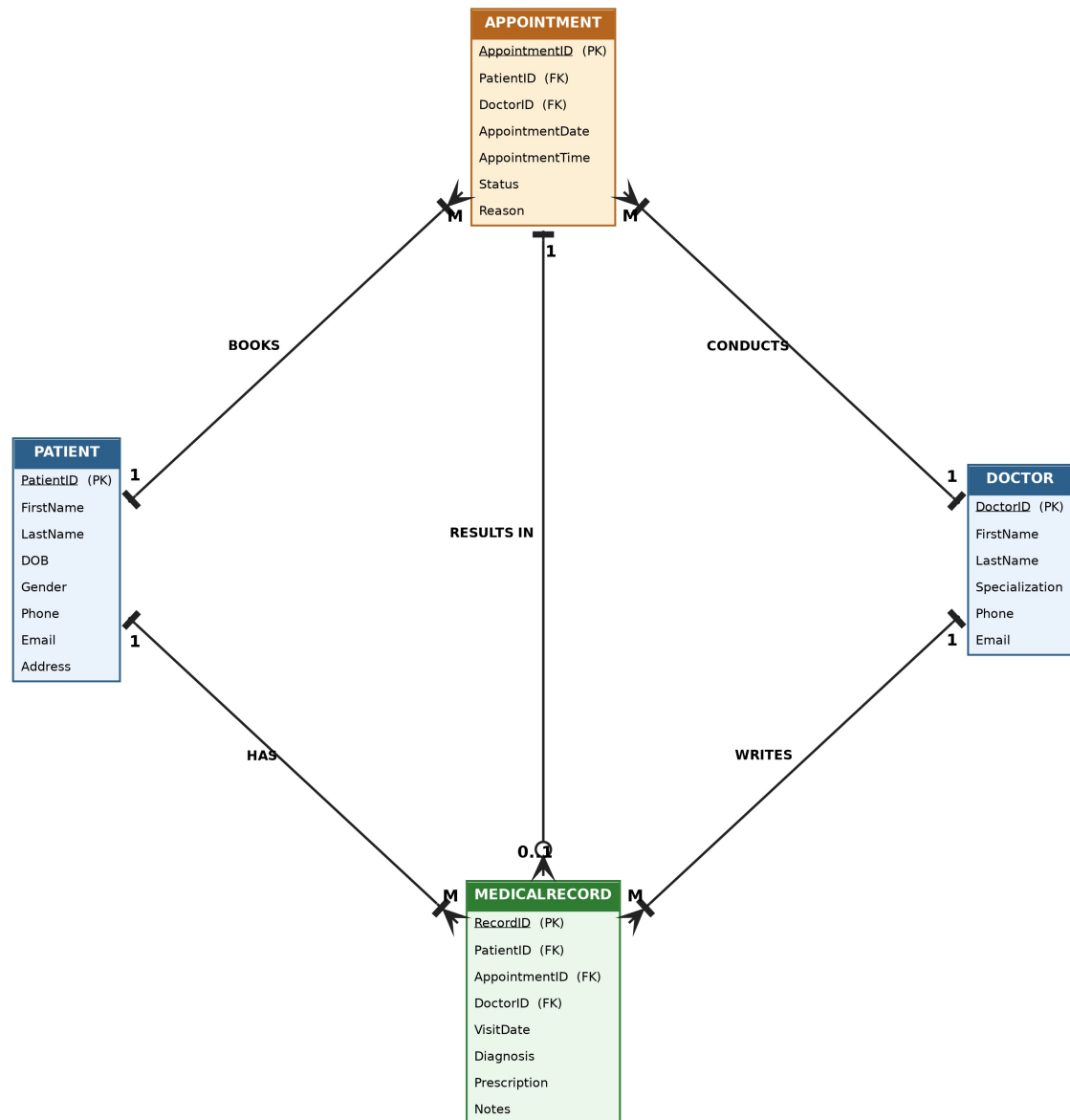
3.1.2 Relationships, Cardinality, and Optionality

Relationship	Cardinality	Optionality
A Doctor conducts many Appointments	1 : M	Mandatory on Appointment side (an appointment must have exactly one doctor); optional on Doctor side (a doctor may have zero appointments)
A Patient books many Appointments	1 : M	Mandatory on Appointment side; optional on Patient side
An Appointment results in at most one MedicalRecord	1 : 0..1	Optional — an appointment only produces a record once the visit is completed
A Patient has many MedicalRecords	1 : M	Mandatory on MedicalRecord side; optional on Patient side
A Doctor writes many MedicalRecords	1 : M	Mandatory on MedicalRecord side; optional on Doctor side

3.1.3 ERD Diagram — Crow's Foot Notation

The diagram below uses standard **crow's foot notation** and is laid out as a diamond (Appointment at top, Doctor at right, MedicalRecord at bottom, Patient at left) so that **no two connecting lines cross**. Each line carries its relationship name in the middle (e.g. "BOOKS", "CONDUCTS") and an explicit **cardinality number at both ends** (1, M, or 0..1) right next to the crow's-foot symbol itself, so the multiplicity of every relationship is readable at a glance: a single perpendicular bar (—|—) means "one and only one," a bar crossed with a crow's-foot (—|<) means "one or many" (mandatory many), and a circle crossed with a crow's-foot (—o<) means "zero or many/one" (optional). Reading the *Appointment* ↔ *MedicalRecord* line, for example: one Appointment results in **zero or one (0..1)** MedicalRecord (a record only exists once the visit is completed), while every MedicalRecord must trace back to **exactly one (1)** Appointment.

Figure 3. Entity-Relationship Diagram (Crow's Foot Notation) — CareConnect Clinic Appointment System



Entity-Relationship Diagram in Crow's Foot Notation

A002	P001	Victoria Okoro	0803-201-4477	D02	Dr. Babatunde Okonkwo	Dermatology	2026-08-02	10:30	Rash
A003	P002	John Okeke	0807-556-2290	D01	Dr. Adaeze Okafor	Cardiology	2026-07-21	09:30	Follow-up

This is now in 1NF: every cell is atomic and every row is uniquely identified by `AppointmentID`.

3.2.3 Second Normal Form (2NF)

Rule: Must be in 1NF, and every non-key attribute must depend on the **whole** primary key, not just part of it (relevant when the key is composite).

Here the primary key `AppointmentID` is a single column, so no attribute can be *partially* dependent — 2NF is technically already satisfied. However, we still separate out the attributes that depend on `PatientID` alone (not on the appointment) and on `DoctorID` alone (not on the appointment), because they are clearly facts about the patient/doctor rather than about the specific appointment. This step anticipates the 3NF fix and is standard practice: we split into three tables:

Appointment (`AppointmentID` PK, `PatientID` FK, `DoctorID` FK, `ApptDate`, `ApptTime`, `Reason`) **Patient** (`PatientID` PK, `PatientName`, `Phone`) **Doctor** (`DoctorID` PK, `DoctorName`, `Specialization`)

3.2.4 Third Normal Form (3NF)

Rule: Must be in 2NF, and no non-key attribute may depend on another non-key attribute (no transitive dependencies).

In the split tables above, `Specialization` depends on `DoctorName` / `DoctorID` (a transitive-looking dependency was already removed by isolating `Doctor` into its own table), and `Phone` depends only on `PatientID` (already isolated). Checking each resulting table:

- **Patient**(`PatientID` PK, `PatientName`, `Phone`) — `PatientName` and `Phone` depend only on `PatientID`. ✓ 3NF.
- **Doctor**(`DoctorID` PK, `DoctorName`, `Specialization`) — both attributes depend only on `DoctorID`. ✓ 3NF.
- **Appointment**(`AppointmentID` PK, `PatientID` FK, `DoctorID` FK, `ApptDate`, `ApptTime`, `Reason`) — all attributes depend only on `AppointmentID` (or are foreign keys), with no attribute depending on another non-key attribute. ✓ 3NF.

The final 3NF schema (extended with `MedicalRecord`, which follows the same reasoning — its attributes `Diagnosis`, `Prescription`, and `Notes` depend only on `RecordID`) eliminates all update/insertion/deletion anomalies: a doctor's specialization is edited in exactly one row, a patient's phone number is edited in exactly one row, and appointments simply reference patients and doctors by ID.

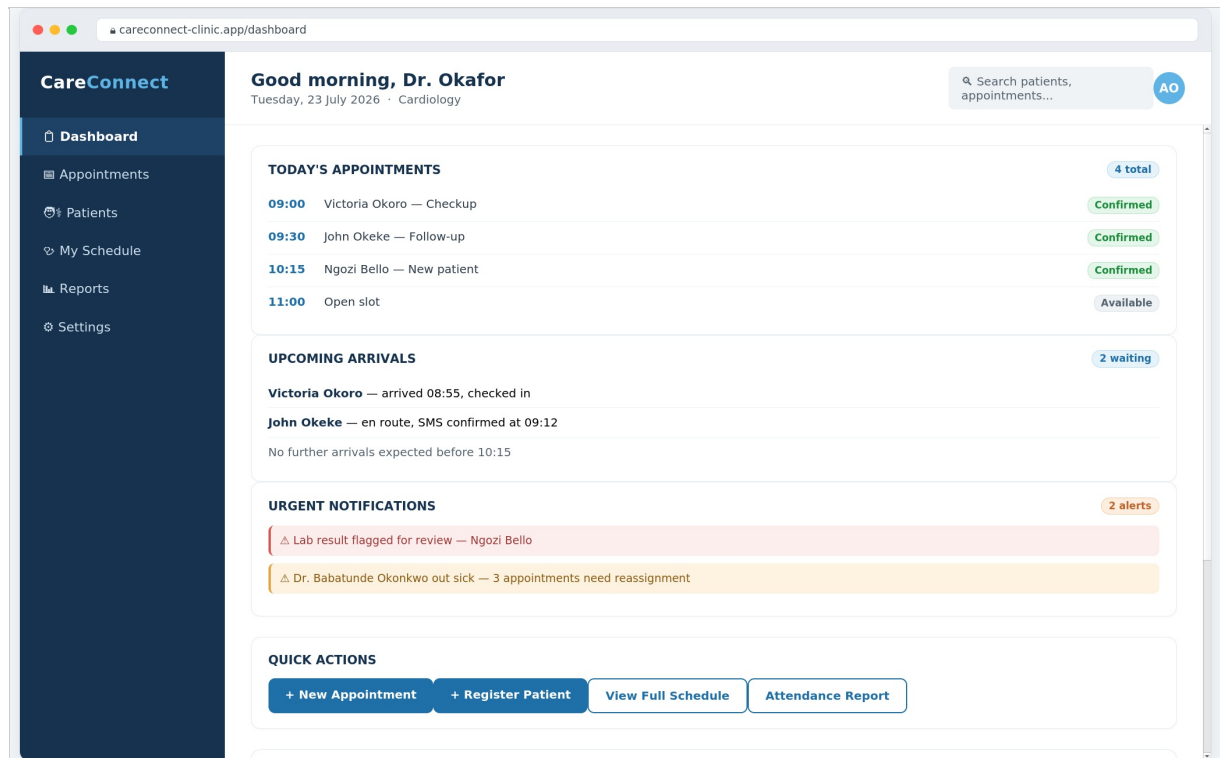
Deliverable produced above: ERD (Section 3.1) and step-by-step normalization to 3NF (Section 3.2).

Step 4 — System Design & Interface Prototyping

4.1 User Interface (UI) Design

4.1.1 Dashboard (Doctor/Staff)

The dashboard is the landing screen after login and surfaces the three things staff need at a glance: **today's appointments**, **upcoming patient arrivals** (who has checked in or is en route), and **urgent notifications** (e.g. a flagged lab result or a doctor calling in sick, which affects other patients' bookings). Quick-action buttons for the two most common tasks — creating an appointment and registering a patient — are always visible.



Doctor/Staff Dashboard Mockup

4.1.2 Data Entry Form (New Patient Registration / Appointment Booking)

Required fields are marked with an asterisk and validated inline: **Full Name**, **Date of Birth** (must be a valid past date), **Phone Number** (10-digit numeric format), and a **Preferred Date/Time** picker that only displays the doctor's genuinely open slots (server-side availability check prevents double-booking, satisfying FR-4). Email and address are optional but validated for format if entered. The Submit button stays disabled until all required fields pass validation, reducing bad data entering the database.

CareConnect

Patient Portal

Hi, Victoria Okoro

1 Patient Details

2 Choose Doctor & Time

3 Confirm

New Patient Registration

Fields marked * are required. Your information is encrypted and only visible to your care team.

Full Name *

Victoria N. Okoro

✓ Looks good

Date of Birth *

14 / 03 / 1991

✓ Valid date

Gender

Female

Phone Number *

0803-201-4477

✓ Valid format

Email

victoria.okoro@email.com

Preferred Doctor

Dr. Adaeze Okafor — Cardiology

Reason for Visit

Routine checkup

Preferred Date & Time * — Tuesday, 23 July 2026

09:00 AM

09:30 AM

10:15 AM

11:00 AM

11:30 AM

01:00 PM

01:30 PM

02:00 PM

Only the doctor's genuinely open slots are shown — booking a taken slot is not possible.

⚠ You will receive an SMS and email reminder 24 hours before your appointment.

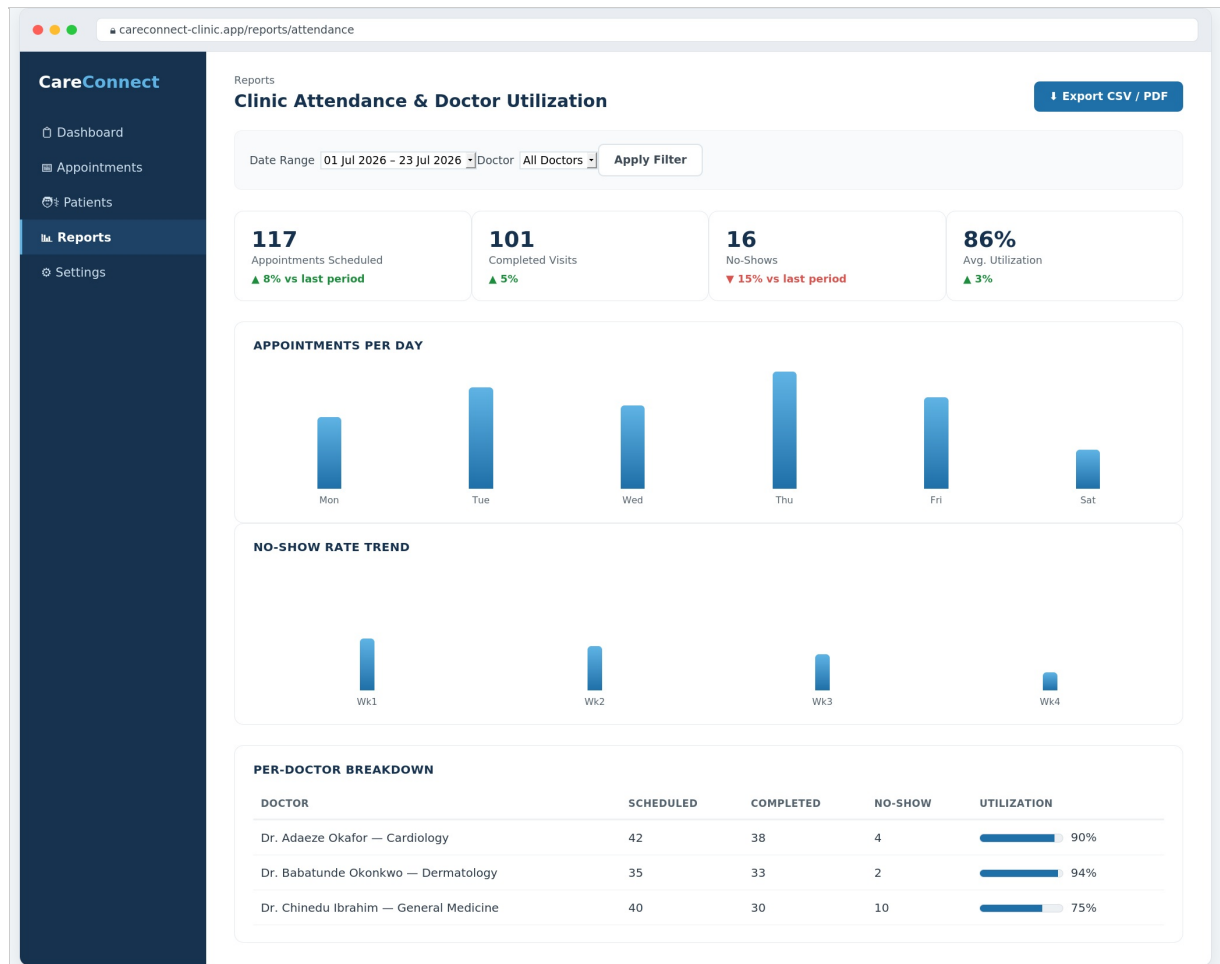
Back

Continue to Confirm →

Registration / Booking Form Mockup

4.1.3 Report Screen (Clinic Attendance Statistics / Doctor Utilization)

The report screen lets the Clinic Manager filter by date range and doctor, view summary charts (appointments per day, no-show trend), and see a per-doctor breakdown of scheduled vs. completed vs. no-show appointments with a computed utilization percentage. Results can be exported to CSV/PDF for offline review or board reporting.



Report Screen Mockup

4.2 Database Schema

Based on the normalized ERD from Step 3, the final schema is:

Table: Patient | Column | Data Type | Key | |—|—|—| | PatientID | INT, AUTO_INCREMENT | PK | | FirstName | VARCHAR(50) | | LastName | VARCHAR(50) | | DOB | DATE | | Gender | VARCHAR(10) | | Phone | VARCHAR(15) | | Email | VARCHAR(100) | | Address | VARCHAR(255) | |

Table: Doctor | Column | Data Type | Key | |—|—|—| | DoctorID | INT, AUTO_INCREMENT | PK | | FirstName | VARCHAR(50) | | LastName | VARCHAR(50) | | Specialization | VARCHAR(50) | | Phone | VARCHAR(15) | | Email | VARCHAR(100) | |

Table: Appointment | Column | Data Type | Key | |—|—|—| | AppointmentID | INT, AUTO_INCREMENT | PK | | PatientID | INT | FK → Patient.PatientID | | DoctorID | INT | FK → Doctor.DoctorID | | AppointmentDate | DATE | | AppointmentTime | TIME | | Status | VARCHAR(20) | (e.g. Scheduled, Completed, Cancelled, No-show) | | Reason | VARCHAR(255) | |

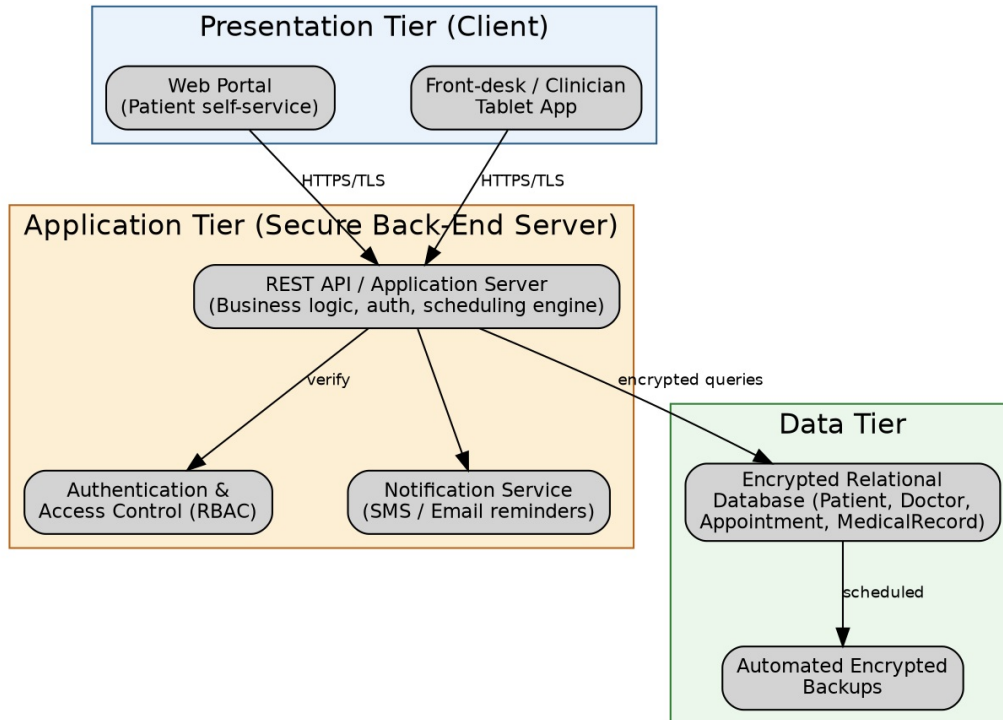
Table: MedicalRecord | Column | Data Type | Key | |—|—|—| | RecordID | INT, AUTO_INCREMENT | PK | | PatientID | INT | FK → Patient.PatientID | | AppointmentID | INT | FK → Appointment.AppointmentID | | DoctorID | INT | FK → Doctor.DoctorID | | VisitDate | DATE | | Diagnosis | TEXT | | Prescription | TEXT | | Notes | TEXT | |

4.3 System Architecture

CareConnect adopts a **secure 3-tier client-server architecture**:

- **Presentation tier (client):** a responsive web portal for patients, and a tablet/desktop application for front-desk staff and clinicians. All communication uses HTTPS/TLS.
- **Application tier (secure back-end server):** a REST API housing the business logic — scheduling engine (conflict/double-booking checks), authentication and role-based access control (RBAC) so patients, nurses, doctors, and managers only see what their role permits, and a notification service for SMS/email reminders.
- **Data tier:** an encrypted relational database holding Patient, Doctor, Appointment, and MedicalRecord tables, with automated encrypted backups running on a daily schedule to meet the 24-hour RPO defined in NFR-7.

Figure 5. Proposed System Architecture — Secure 3-Tier Client-Server Model



System Architecture Diagram

Deliverable produced above: UI mockups (Section 4.1), database schema (Section 4.2), and system architecture (Section 4.3).

Step 5 — System Implementation & Review

5.1 System Specification Document (Consolidated)

This document consolidates all prior work into a single blueprint for the CareConnect Clinic Appointment System:

- **Project Charter & Feasibility Study** — Section 1 (problem statement, goals, scope, stakeholders, and technical/economic/operational feasibility).
- **Requirements** — Section 2 (10 user stories, 10 functional requirements, 7 non-functional requirements).
- **Data Flow Diagrams** — Section 2.4 (Context-level and Level-0 DFDs).
- **Entity-Relationship Diagram & Normalization** — Section 3 (4 core entities, cardinalities, and a full 1NF→2NF→3NF walkthrough).
- **UI Designs, Database Schema & Architecture** — Section 4 (dashboard, registration/booking form, and report screen mockups; final schema for 4 tables; 3-tier secure architecture).

Together these sections form a complete, implementation-ready blueprint: a development team could take Sections 3–4 directly into database creation and application scaffolding, while Sections 1–2 provide the business justification and requirements traceability needed for sign-off.

5.2 Final Presentation Outline (10–15 minutes)

A presentation to CareConnect Clinic Management should follow this structure:

Time	Segment	Content
0:00–2:00	The Problem	Current manual-process pain points: scheduling conflicts, lost records, long wait times, administrative burden (Section 1.2).
2:00–4:00	The Proposed Solution	High-level overview of the CareConnect system and its scope (what it will and will not do).
4:00–7:00	Live Walkthrough via UI Mockups	Demonstrate the patient booking flow, the doctor/staff dashboard, and the manager’s report screen (Section 4.1) to show exactly how each stakeholder’s day-to-day work changes.
7:00–10:00	Key Technical Decisions	Briefly explain the 3-tier secure architecture (Section 4.3), the normalized database design (Section 3.2), and why these choices support reliability and growth.
10:00–13:00	Patient Privacy & Workflow Efficiency	Explain encryption, role-based access, and audit logging (NFR-1, NFR-5) alongside the projected reduction in no-shows and administrative time (Section 1.6.2 cost-benefit).
13:00–15:00	Benefits Recap & Next Steps	Restate the cost-benefit conclusion (positive ROI within Year 1), propose a phased/pilot rollout, and open the floor for questions.

Deliverable produced above: the complete “CareConnect System Specification Document” (this document, Sections 1–5) and the presentation outline above, ready to be adapted into slides.